

Neuromuscular electrical stimulation in patients with heart failure*

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April 23, 2020

Abstract

Evidence establishing the effectiveness of neuromuscular electrical stimulation (ES) in patients with heart failure (HF) is growing with five systematic reviews published within the past eight years and two recent moderately sampled well-controlled trials already published in 2017. This paper reviews this evidence and develops an empirically based yet rationally derived causal model of ES for physical therapy practice considerations when treating patients with HF. The paper develops a graphic causal model using prior knowledge regarding the physiology of HF, aerobic capacity, endurance and quality of life combined with available empirical evidence on the NMES intervention in patients with HF. The paper reviews reasoning patterns based on the causal structure during intervention-based reasoning and inter causal consequences (confounding and biasing tendencies). Finally, the paper develops reasonable decision trees (algorithms) based on the strengths of the causal structure and develops suggestions for future clinical trials.¹

1 Introduction²

The causal relationship evaluated in this paper reflects interventional reasoning. The intervention being reviewed is the use of neuromuscular electrical stimulation (ES) in patients with heart failure (HF). Several causal models are developed and merged incrementally and parameterized. The first model (Model 1) is on the physiology of heart failure, aerobic capacity, and endurance. The second model (Model 2) is on determinants of quality of life in patients with heart failure. These two models will be merged into what could be considered an underlying hierarchical (or emergent) causal model (Model 3). The fourth model (Model 4) starts with Model 3 and adds interventions. This addition is

*Draft - work in progress; this version is an example KBP project for Clinical Inquiry I[?]

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¹Either of these last steps are optional for the Clinical Inquiry I

²This paper was prepared in L^AT_EX for typesetting in L^AT_EX and is available on GitHub at https://github.com/scollinspt/clinical_inquiry

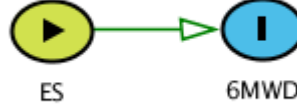


Figure 1:

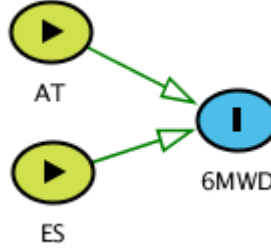


Figure 2:

based on empirical evidence on how and to what extent neuromuscular electrical stimulation (ES) has been shown to influence aerobic capacity, endurance and quality of life in patients with HF. While there is a growing body of evidence for this intervention,[?, ?, ?, ?, ?, ?, ?] this draft focuses on one systematic review.[?]

The systematic review by Gomes et al[?] summarizes the literature that tests the questions:

1. Does ES, as compared to a sedentary control, increase maximal oxygen consumption ($\dot{V}O_2\text{Max}$)?
2. Does ES, as compared to a sedentary control, increase six minute walk distance (6MWD)?
3. Does ES, as compared to a sedentary control, improve quality of life (QOL)?
4. Does ES, as compared to aerobic training (AT), increase maximal oxygen consumption ($\dot{V}O_2\text{Max}$)?
5. Does ES, as compared to AT, increase six minute walk distance (6MWD)?
6. Does ES, as compared to AT, improve quality of life (QOL)?

This paper will focus on questions 1 - 3. Figure 1 depicts a sample causal model of question 2, and Figure 2 depicts a sample causal model of question 5.

KBP[?]considers interventional reasoning based on causal knowledge, and considers causal knowledge probabilistic. We can depict interventional statements as graphs (Figure 1 and 2), and we can also consider what these mean

based on conditional probability. Note that the below conditional probability formulations allow precise and explicit statements of what a hypothesis is testing (a causal assumption which is an edge drawn in a causal model), and is exactly what a clinician wants to know about an intervention. That is, what is the probability of Y_1 (some improvement of a certain level in an important variable related to the client's goals) given use of some intervention is performed (X_1).

Just a test!

- **Hypothesis 2 (based on question 2)**

- $P(6MWD_1 | ES_1) \geq P(6MWD_1 | ES_0)$, which means the probability of 6MWD increases ($6MWD_1$) given we use ES (ES_1) is greater than the $6MWD_1$ given we do not use ES (ES_0)

- **Hypothesis 5 (based on question 5)**

- $P(6MWD_1 | ES_1, AT_0) \geq P(6MWD_1 | ES_0, AT_1)$, which means the probability of 6MWD increases ($6MWD_1$) given we use ES (ES_1) and do not use AT (AT_0) is greater than the $6MWD_1$ given we do not use ES (ES_0) and we use AT (AT_0)

2 Rational Conceptions: Underlying model

2.1 Model 1: Core underlying physiology

Figure 3 includes Model 1, the core underlying physiology. This model is based on the knowledge that the author proposes mechanistically links the intervention to the outcomes measured. This model is based on well-understood physiological concepts. For example, the intervention (ES) is proposed to change the physiology of muscle. Muscle physiology influences aerobic capacity and endurance.

2.2 Model 2: Quality of life[?]

Krethong et al[?] propose that health related quality of life (HRQOL) in patients with heart failure is based on a set of variables as depicted in Figure 4.

2.3 Model 3: Underlying Model of Heart Failure, Aerobic Capacity and Quality of Life

This model combines Model 1 and 2 with the following assumptions: biological / physiological status is dependent on $\dot{V}O_2\text{Max}$ and Functional Status is dependent on 6MWD as depicted in Model 3 (Figure 5). models. and a model 2 will be based on a paper on the determinants of quality of life in patients with heart failure.

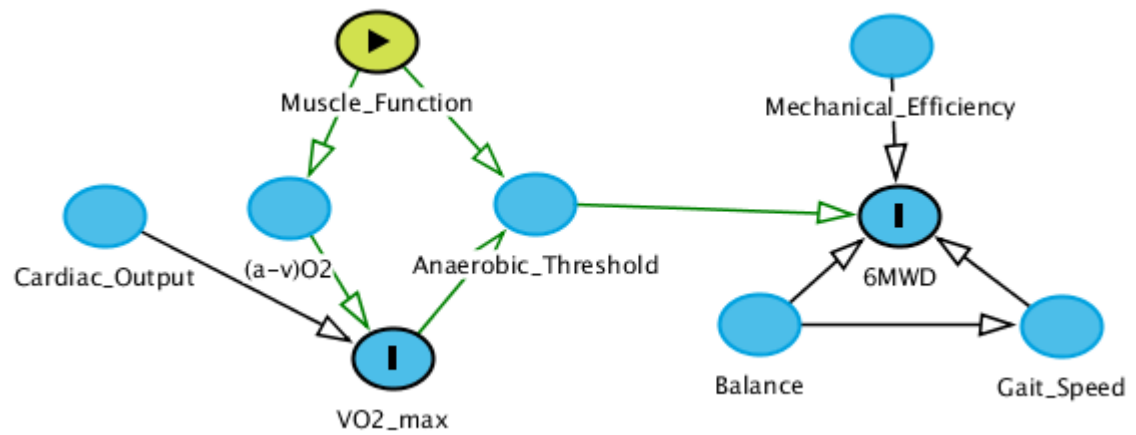


Figure 3:

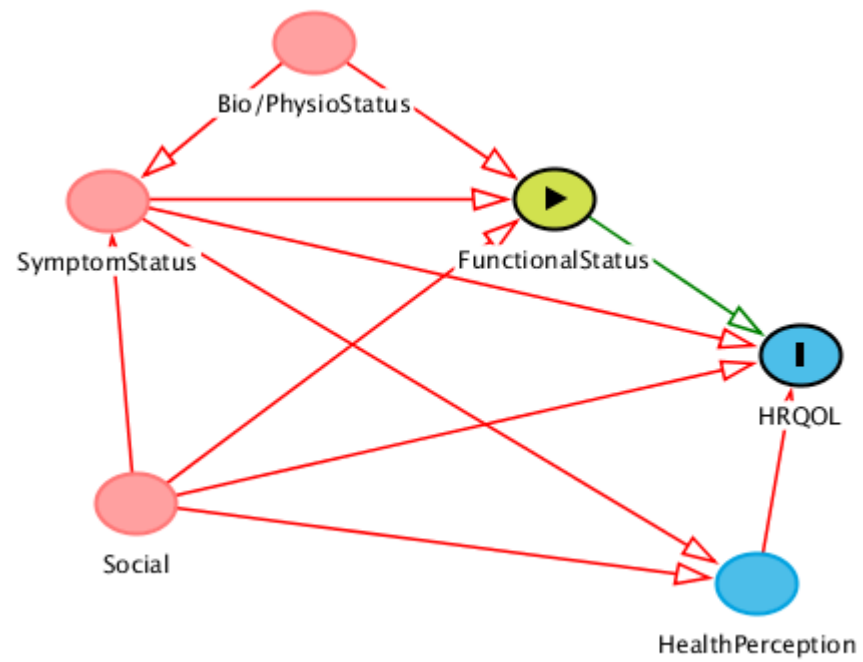


Figure 4:

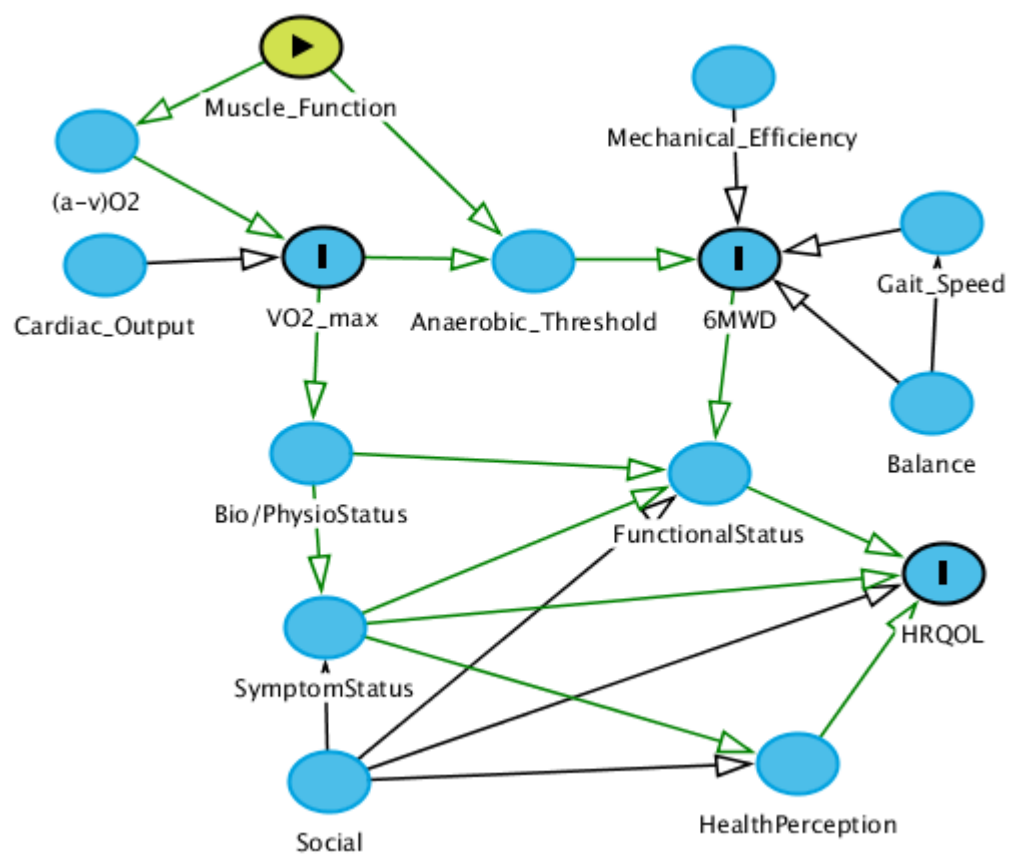


Figure 5:

3 Empirical Evidence: Graphical causal model of intervention effects

Empirical evidence on the intervention effect of ES are drawn from the systematic review and meta-analysis by Gomes-Neto et al.[?] The focus for this paper are hypotheses 1 - 3 from the introduction, restated here:

- $P(\dot{V}O_2\text{Max}_1 | ES_1) \geq P(\dot{V}O_2\text{Max}_1 | ES_0)$
- $P(6\text{MWD}_1 | ES_1) \geq P(6\text{MWD}_1 | ES_0)$
- $P(\text{HRQOL}_1 | ES_1) \geq P(\text{HRQOL}_1 | ES_0)$

The empirical evidence is presented in the meta-analysis as a test of the hypothesis that there is a statistically significant difference in the effect size between the ES intervention groups and the control groups in the reviewed studies.

3.1 Table 1. Effect Sizes for Primary Outcomes

Hypothesis	Effect Size [95% CI]
$P(\dot{V}O_2\text{Max}_1 ES_1) \geq P(\dot{V}O_2\text{Max}_1 ES_0)$	1.85 [0.46, 3.23]
$P(6\text{MWD}_1 ES_1) \geq P(6\text{MWD}_1 ES_0)$	63.54 [35.81, 91.21]
$P(\text{HRQOL}_1 ES_1) \geq P(\text{HRQOL}_1 ES_0)$	0.82 [0.10, 1.53]

3.2 Conditional Probabilities

3.2.1 Approach

While effect sizes are useful, they are statements of the sample mean and do not represent the conditional probability of improvement (or change) in an outcome (effect) given the presence of the intervention (cause). It is this conditional probability that is important to the clinician, and embedded within a probabilistic graphical model. The conditional probabilities calculated in Table 2 are based on the following calculations, and are based on assumptions included in Section 3.2.3.

3.2.2 Calculations³

The paper provides the upper (and lower) bound of the confidence interval for the estimated effect. We can use this to determine the standard deviation (σ).

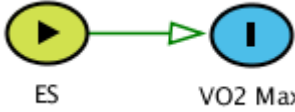
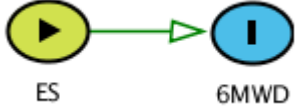
- $CI_{upper} = \bar{x} + 1.96 \left(\frac{\sigma}{\sqrt{n}} \right)$ (equation for upper bound of CI)
- $\sigma = \frac{CI_{upper} - \bar{x}}{1.96} \cdot \sqrt{n}$ (solving for σ)

³There is an excel spreadsheet calculator for these calculations at: https://github.com/scollinspt/clinical_inquiry/tree/master/Clinical_InquiryIprojectFiles

With σ , and assuming a normal distribution, we can calculate the probability that Y falls within a certain improvement (denoted by the minimally important clinical difference (MICD)) with this equation.

- $P(Y_1|X_1) = \left(\frac{\bar{x} - MICD}{\sigma} \cdot 0.34 \right) + 0.50$

3.2.3 Table 2

Conditional Probability	Simplified Graphic Depiction	
$P(\dot{V}O_2Max_1 ES_1)$		0.51 (MICD of 1.6)
$P(6MWD_1 ES_1)$		0.56 (MICD of 30.1)
$P(HRQOL_1 ES_1)$		0.64 (MICD of 0)

4 Graphical Causal model of NMES in Heart Failure

4.1 Model 4 (non parameterized - DAGitty)

Model 4 (Figure 6) simply adds the intervention (ES) to the underlying model as a non parameterized model of the intervention. The decision was made to connect ES to its proposed direct effect, Muscle Function, as opposed to connecting directly to the measured outcomes ($\dot{V}O_2Max$, 6MWD, HRQOL). This is a potentially controversial decision that favors the rationale conceptual model over the empirical evidence. The author believes casting the intervention in this framework provides a better explanation for the resulting measured effects.

4.2 Model 4 (parameterized - SAMIAM)

Model 4 parameterization is based on the empirical evidence and in this draft it is based on the evidence in Gomes Neto et al[?]so requires a simplified format of the model. The empirical studies have simply tested the direct hypotheses without consideration for the intermediary variables presented in the full Model 4 in Figure 6. The SAMIAM model in Figure 7 is cut down to the variables tested and under the assumption of independence between the set of outcome

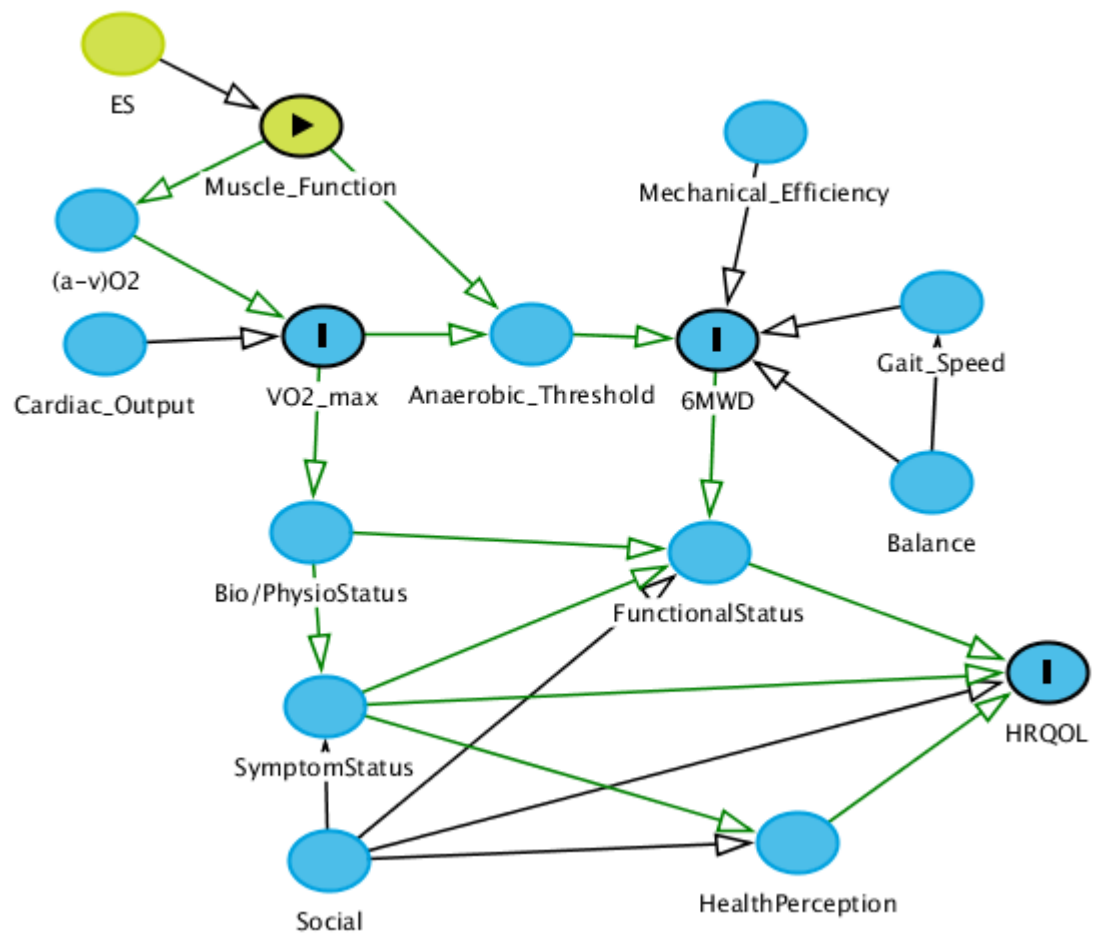


Figure 6:

Figure 7:

variables. The model is shown with ES instantiated to having been provided (ES is presented here as having been provided or not provided (binary variable)).

5 Reasoning Patterns

This section of the paper identifies various forms of reasoning that can be accomplished with the completed parameterized and non-parameterized models.

5.1 Inferences and Causal Paths for interventional reasoning

The empirical evidence supports hypotheses 1 - 3 and we can see that ES (compared to no intervention) provides slightly better than a 50/50 probability for a minimally clinical important difference (MICD) change in $\dot{V}O_2\text{Max}$ and 6MWD; and a change in HRQOL.⁴ It is possible that 6MWD and HRQOL have a greater probability to increase due to being more closely associated with reversible components of physical function and less directly impacted by limited cardiac output. For example, based on Model 4 we see that $\dot{V}O_2\text{Max}$ is more proximally related to cardiac output. It is also possible that improvements in Muscle Function may increase mechanical efficiency, balance or gait speed which would result in an improvement in 6MWD independent of $\dot{V}O_2\text{Max}$. These possible connections are not included in Model 4 but a review of additional literature may be warranted to consider encoding these causal assumptions.

Improvements in HRQOL are possible through several mechanisms. The increased probability of seeing an improvement in HRQOL may be an example of a multiplier effect as the impact of changes associated with ES propagate through the causal network. For example, there are six possible causal paths from ES through $\dot{V}O_2\text{Max}$ and 6MWD to HRQOL.

5.2 Intercausal Paths (bias, confounding)

Confounding pathways can be identified from causal models and fall into several categories. The simplest to identify (and the only one we will focus on for now - in this draft) is a fork. A fork is a cause of both a cause and an effect (variable Z in Figure 8).⁵ Forks are important for both research and clinical practice. In clinical practice knowing about forks is important with both interventional and prognostic (deductive) and diagnostic (abductive) inferences. If we are

⁴Note - this is an assumption regarding the null value being 50/50, I do not think this is an appropriate null value so am working on justification for an alternative

⁵For a concrete example from the literature, see this post: <https://knowledgebasedpractice.wordpress.com/2015/04/21/le-power-and-falls-a-dag-and-adjustment-set-sample-from-recent-literature/>

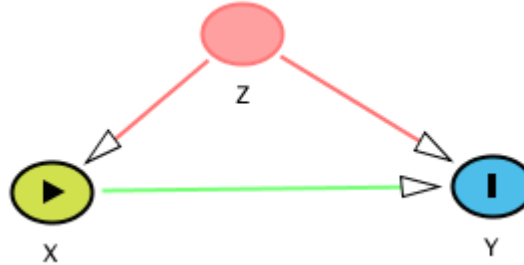


Figure 8:

interested in whether our intervention (X) will result in an outcome (Y) we need to be aware of Z; or if we are interested whether the presence of a clinical sign (Y) says about the probability of the existence of a disease (X) then we also need to be aware of Z.

5.2.1 Examples of confounding paths from Model 4

Model 4 has several confounding pathways to be considered. Figure 9 includes some marked in red.

1. Muscle function fork: Ironically, the systematic review reported in this study did not include Muscle Function as an outcome. There was not enough literature reporting on this variable to include it in the review. Muscle function should not be a confounder as it should be measured and considered when the intervention is primarily aimed at improving muscle function.
2. Balance fork: anytime 6MWD is considered as a variable, gait speed and subsequently balance needs to be considered.
3. Bio/Physio Status fork: this fork may represent one reason that HRQOL has a greater probability of improvement than $\dot{V}O_2\text{Max}$, the impact of symptom status may contribute to the multiplier effect mentioned earlier.
4. Symptom Status fork: another possible contributor to the multiplier effect

6 Algorithms (Optional)

If we focus on the empirical evidence the algorithm is pretty simple. If HF, then ES. Of course, a slightly more sophisticated algorithm could be developed based on knowledge encoded in Model 4 based on information from other variables. For example, we can include If / Then rules based on client characteristics such as whether there are reductions in muscle function (If reduced muscle function, then ES).

This section is still in development - but includes a new section on study characteristics

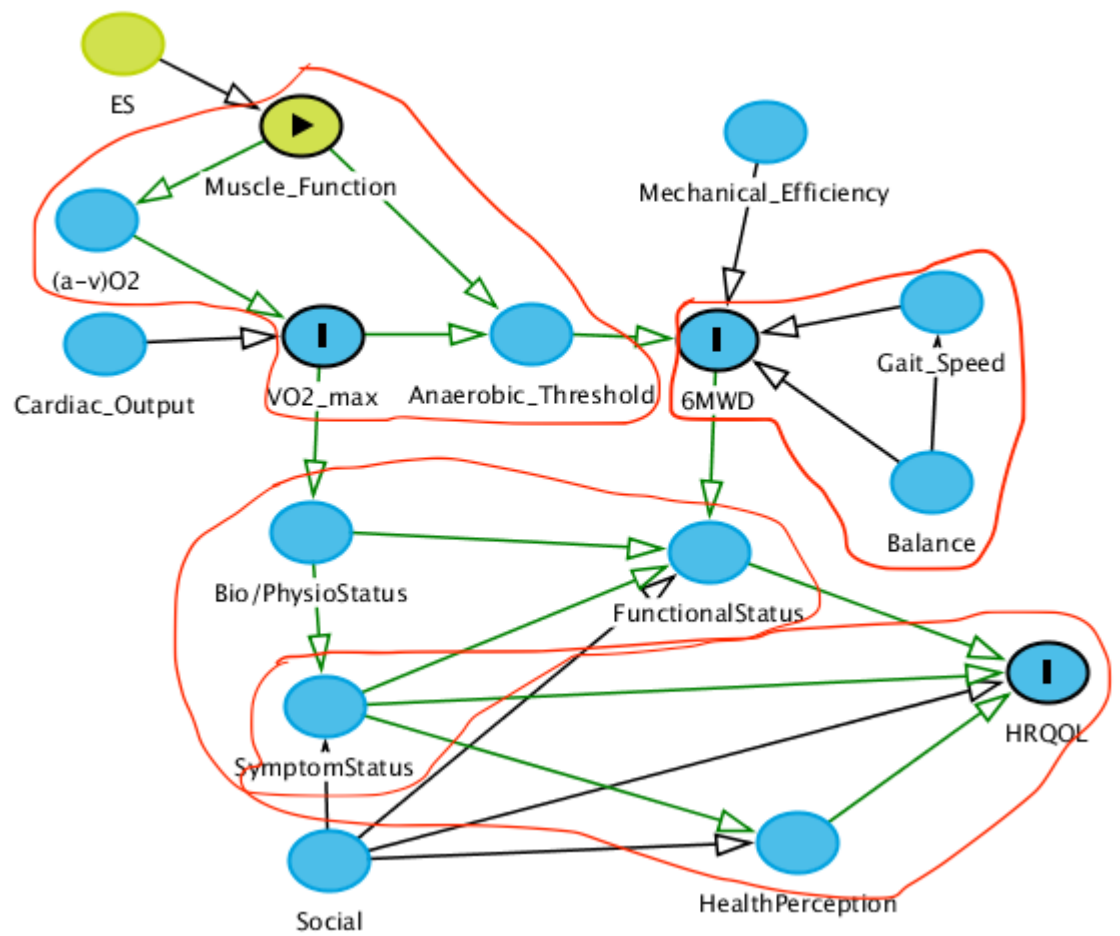


Figure 9:

6.1 Study Characteristics

Once the case for an intervention has been made by a set of studies then study characteristics become the important for algorithm development, or at least for simple considerations such as, who has this intervention been shown to work and under what conditions.

6.1.1 Patient Characteristics

Gomes Neto et al[?]

Sex one study was only males, most included both sexes but there was a predominance of male subjects

Age 50 - 65 (means)

Setting Outpatient

Status NYHA Functional Classes II - IV, all optimized medically

6.1.2 Intervention Characteristics

Gomes Neto et al[?]

Intensity From visible muscle contractions, to 60 mA, to maximum tolerated up to 25-35% of maximal voluntary contraction

Current Predominantly biphasic

Frequency 10 - 50 Hz

Muscles All included quadriceps; some included quadriceps and calf muscles; others included quadriceps and gluteal muscles; some included quadriceps and hamstrings

Time(min/d) 30 - 120

Frequency(d/wk) 5-7

Duration(wks) 2 - 12

7 Future Study Recommendations (Optional)

Still under development. This section of the paper will develop future study recommendations based on the structure of the graphical model (particularly the intercausal paths) and the ability to justify the parameter estimates.